

- Explain the principles of homeostasis.

(b) Glycogen phosphorylase catalyses the conversion of glycogen to glucose in liver cells. The production of glycogen phosphorylase is coded for by the gene *PYGL*.

Suggest **and** explain why cell signalling by glucagon is likely to be affected in the liver cells of a person with GSDVI.

[3]



- (c) Glycogen synthase catalyses the conversion of glucose to glycogen in liver cells. The production of glycogen synthase is coded for by the gene *GYS2*.

A mutation in *GYS2* leads to a condition called glycogen storage disease type 0 (GSD0) in which glycogen is not formed efficiently.

Suggest what the consequences would be if a person with GSD0 has a meal rich in glucose.

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..... [3]

[Total: 10]

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